

PROJECT EXECUTABLE FILES

Date	15 March 2026	Maximum Marks	3 Marks
Team ID	SWTID-2026-9692	Project Name	AI WeatherWise API

Project Description: AI WeatherWise API is an AI-powered RESTful service designed to provide real-time weather metrics, natural language weather summaries, and tailored clothing or activity recommendations using the Google Gemini API. The backend utilizes a Node.js and Express architecture with a MongoDB database for secure user authentication and data persistence.

STEP 1: SUBMISSION CHECKLIST

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	Yes
7	Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	Yes
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

STEP 2: FILE / FOLDER STRUCTURE

```
ai-weatherwise-api/  
├── src/  
│   ├── config/           # Database and environment configurations  
│   ├── controllers/      # Controller logic for endpoints  
│   ├── models/           # MongoDB models (User, WeatherCache, Log)  
│   ├── routes/           # Express route definitions  
│   ├── services/         # Weather API & Gemini AI integrations  
│   └── app.js             # Express app initialization  
├── tests/                # Unit and integration tests  
├── docs/                 # API Documentation & Architecture Diagrams  
└── .env.example          # Template for environment variables
```

```
| Dockerfile           # Container deployment configuration
| package.json        # Node.js dependencies and scripts
| README.md           # Project setup and user manual
```

STEP 3: DEPLOYMENT / ACCESS DETAILS

Hosted / Deployed URL	https://ai-weatherwise-api.onrender.com
Repository Link	https://github.com/smartbridge-projects/SWTID-2026-9692-AI-WeatherWise-API
Login Credentials (Demo)	Username: demo_evaluator@smartbridge.com Password: EvaluatorPass2026!
Hosting Platform Provider	Render / MongoDB Atlas
Demo Video Link	https://drive.google.com/file/d/demo-walkthrough-swtid-9692/view

STEP 4: RUN INSTRUCTIONS

Pre-requisites:

- Node.js (v18.x or higher) and npm installed
- MongoDB instance (Local or MongoDB Atlas cluster connection string)
- Google Gemini API Key

Step 1: Clone the repository: `git clone https://github.com/smartbridge-projects/SWTID-2026-9692-AI-WeatherWise-API.git`

Step 2: Navigate to project directory and install dependencies: `cd ai-weatherwise-api && npm install`

Step 3: Configure environment variables: `cp .env.example .env` and add your `GEMINI_API_KEY` and `MONGO_URI`.

Step 4: Run database seed script (optional for initial user setup): `npm run seed`

Step 5: Start the server in development mode: `npm start`

Step 6: Test API endpoints locally at `http://localhost:5000/api/v1/health` or refer to Swagger docs at `http://localhost:5000/api-docs`.

STEP 5: KNOWN ISSUES / LIMITATIONS

S.No	Known Issue / Limitation	Workaround / Status
1	Free tier of external weather API has a limit of 60 calls per minute.	Implemented response caching using Redis/MongoDB cache with a 10-minute TTL.
2	Occasional latency during peak Gemini API response generation.	Added loading indicators and fallback standard textual responses.
3	Historical weather data limited to the past 30 days on current plan.	Documented as a future scope upgrade feature.